London Ambulance Service NHS
NHS Trust

Administration of medical gases in ambulance patient care

Component 23

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Medical Gases- Component 23

Objective

- Explain benefits of oxygen therapy
- Identify indications for the use of oxygen
- Explain contraindications prohibiting use of Oxygen
- Describe dangers of using compressed gas
- Outline health and safety rules

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Why do we need oxygen?

- Provides the fuel for every cell in the body
- Needed for the cells to produce the chemical activity to function
- Normal body function maintains our oxygen levels between 94% and 98% saturation on red blood cells

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Definitions

- Hypoxia: inadequate oxygen tension at cellular level

Characterised by; Tachycardia, Hypertension, peripheral vaso-constriction, dizziness and mental confusion

- Hypoxaemia: an abnormal deficiency in the concentration of O2

in arterial blood

Characterised by cyanosis, restlessness, <GCS

CheyneStokes respiration, apnoea, hypertension and tachycardia

- Hypercapnia (aka hypercarbia): greater than normal CO2 in the blood

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Implications and Benefits of Oxygen Therapy

- Oxygen is a medication and should be treated as such
- Should only be prescribed and administered to those with a clinical need
- Oxygen is the treatment for hypoxaemia and not breathlessness
- Oxygen therapy should be titrated to achieve normal saturation values in healthy adults(94-98%)

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Benefits of Oxygen Therapy

By increasing the oxygen content within the lungs you can force the haemoglobin to carry and distribute more oxygen, thereby reversing the effects of hypoxia on the brain, heart and other vital organs.



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Oxygen

- Indications
- Contra-indications
- Cautions
- Side Effects
- Dosage
- Routes of administration



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Oxygen Cylinders

- For ambulance purposes oxygen is stored in cylinders
- Oxygen therapy can be essential in patient care and the equipment must be maintained in good working order
- You must know how and when to operate it correctly

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- In the UK oxygen cylinders are clearly marked with ‘Oxygen’ or ‘O2’ and are charged at 2,000 psi or more
- Lightweight cylinders
- Always check the label

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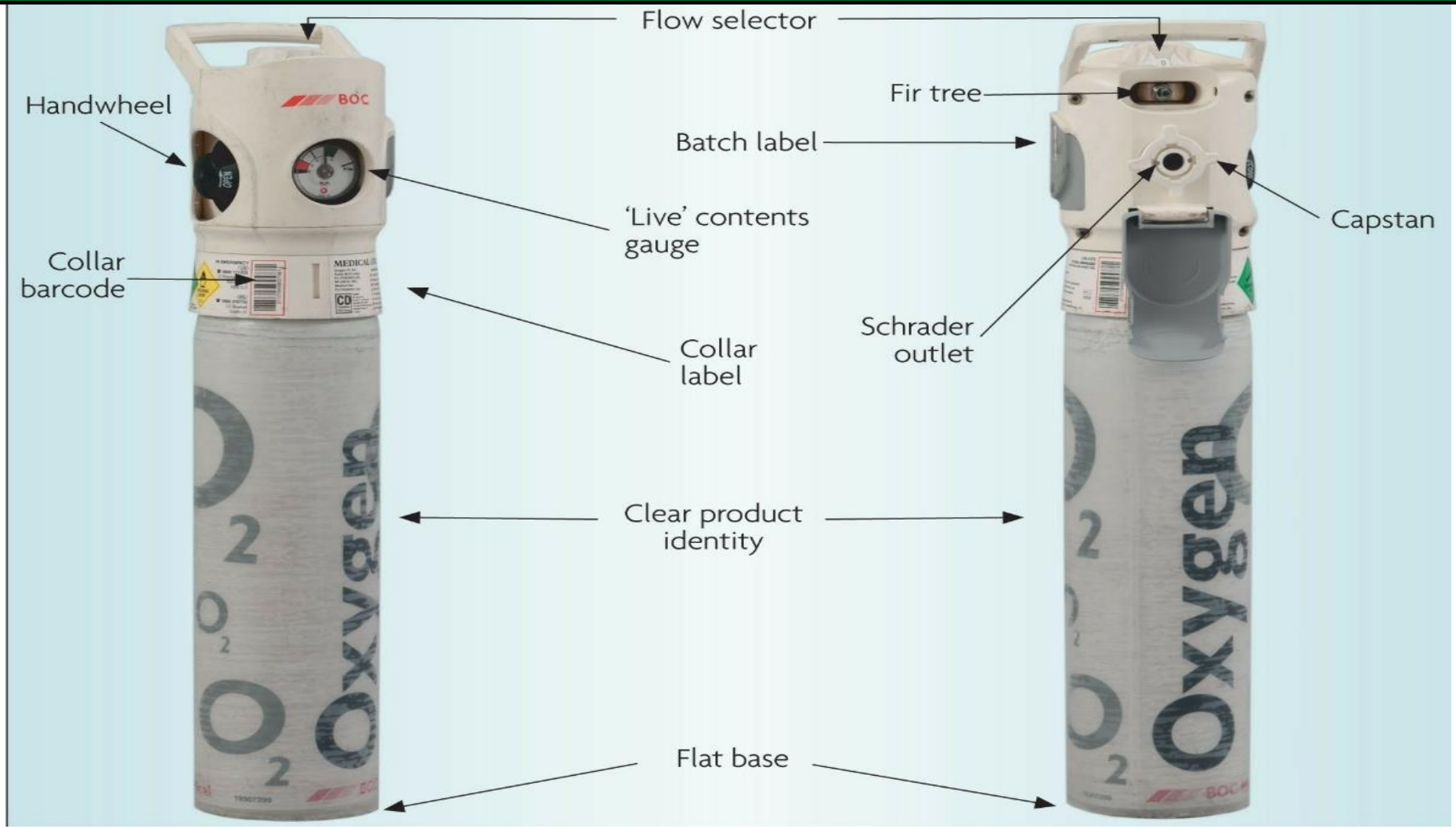
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There are two sizes of oxygen cylinder currently
used within the LAS

- ZD size = 605 litres
- HX size = 2300 litres

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1. Check expiry



2. Open/Close valve



3. Flow regulator

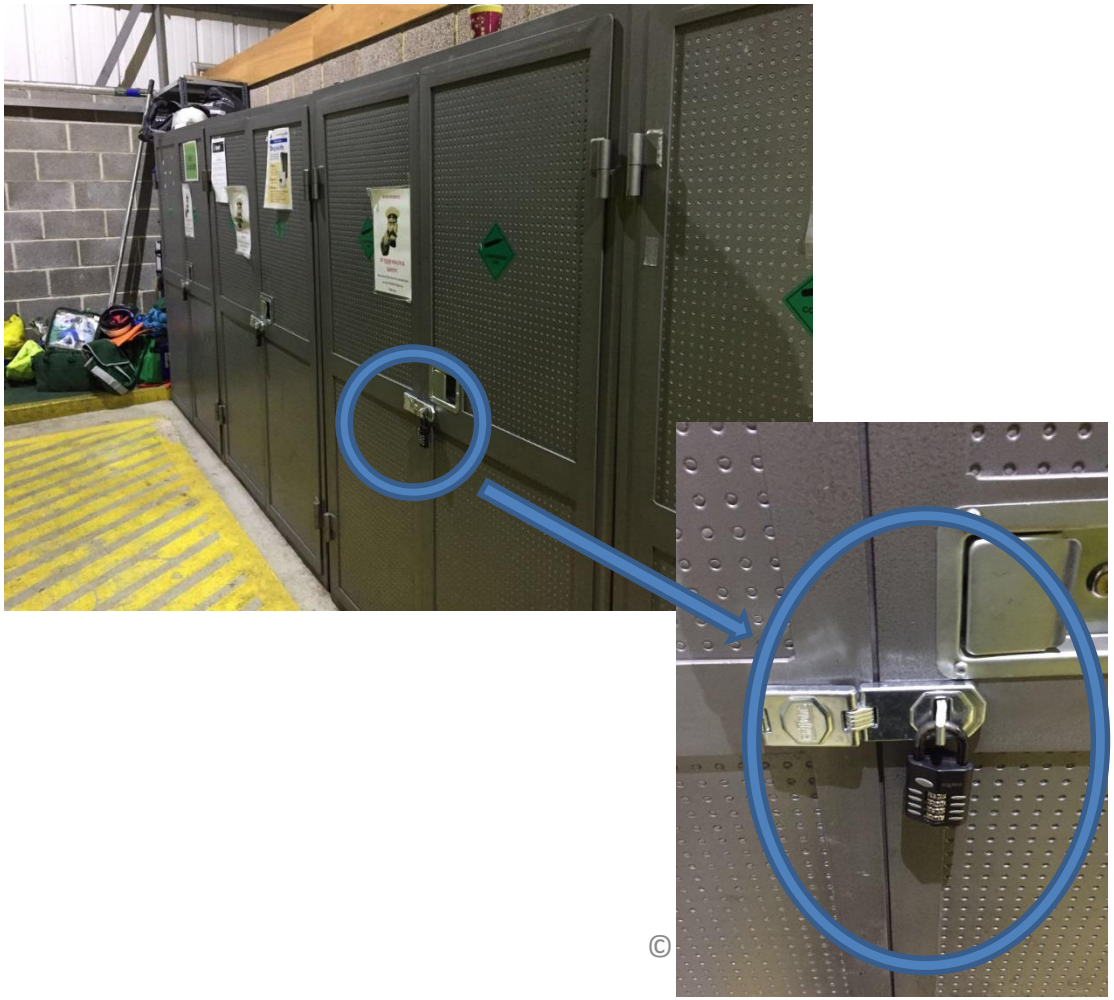


4. Contents gauge (do not use if in the red)

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Storage





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USE WITH CARE

- All materials that are combustible in air are ignited more easily and burn with a fiercer flame in the presence of pure oxygen

- Oil, grease, sparks, flames or any other type of ignition source may also cause an explosion if it comes into contact with oxygen equipment

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Health & Safety

- Medical gas cylinders are subject to strict regulations on storage and safe use
- Cylinders must be stored in well ventilated positions in special areas or cupboards provided
- Full and empty cylinders should be kept separated
- The storage area should not be used for any other purpose
- Cylinders should be properly supported off the floor

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Health & Safety

- Cylinders should not be allowed to get rusty, dirty, be repainted or have their markings obscured
- Cylinders must not be stored near stocks of combustible and / or flammable materials
- Warning signs prohibiting smoking or naked lights must be visible
- Adequate access must be provided and maintained to and from storage area

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Health & Safety

- No smoking when using oxygen
- Do not use in the presence of naked flames
- Exercise care when using oxygen in presence of machinery or equipment which can create sparks
- Oil or grease must not come into contact with the outlet, valves or associated equipment due to risk of explosion

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Health & Safety

- Newly charged cylinders must have a securely fixed seal on outlet
- The outlet face must not be damaged
- Avoid dropping or rolling cylinders along ground when moving them
- Avoid excessive force when turning the valve key or when connecting any apparatus

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Regulators

These are attached to the cylinders to reduce the pressure from 2,000 psi to around 60 psi

They incorporate a contents gauge and a flow meter,
which is attached either directly or at the end of a piping system to assist in varying the flow rate of oxygen being delivered to the patient

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Oxygen - Equipment

Bag-valve mask – 15ltr
Non-rebreather mask – 10 to 15ltr
Simple face mask – 5 to 10ltr
Venturi mask – 4ltr
Nasal Cannula – 2-6ltr
Nebuliser – 6 to 8ltr



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Problems recording oxygen saturation

Heavily applied nail varnish



Consider a quarter turn of the finger probe to make contact with the skin



Poor circulation due to cold, disease or sudden illness

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Entonox



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Objectives

- Describe properties of Entonox
- Explain complications of environmental Temperature
- Explain benefits of Entonox therapy
- Identify the indications for the use of Entonox
- Explain cautions and contraindications for use of Entonox

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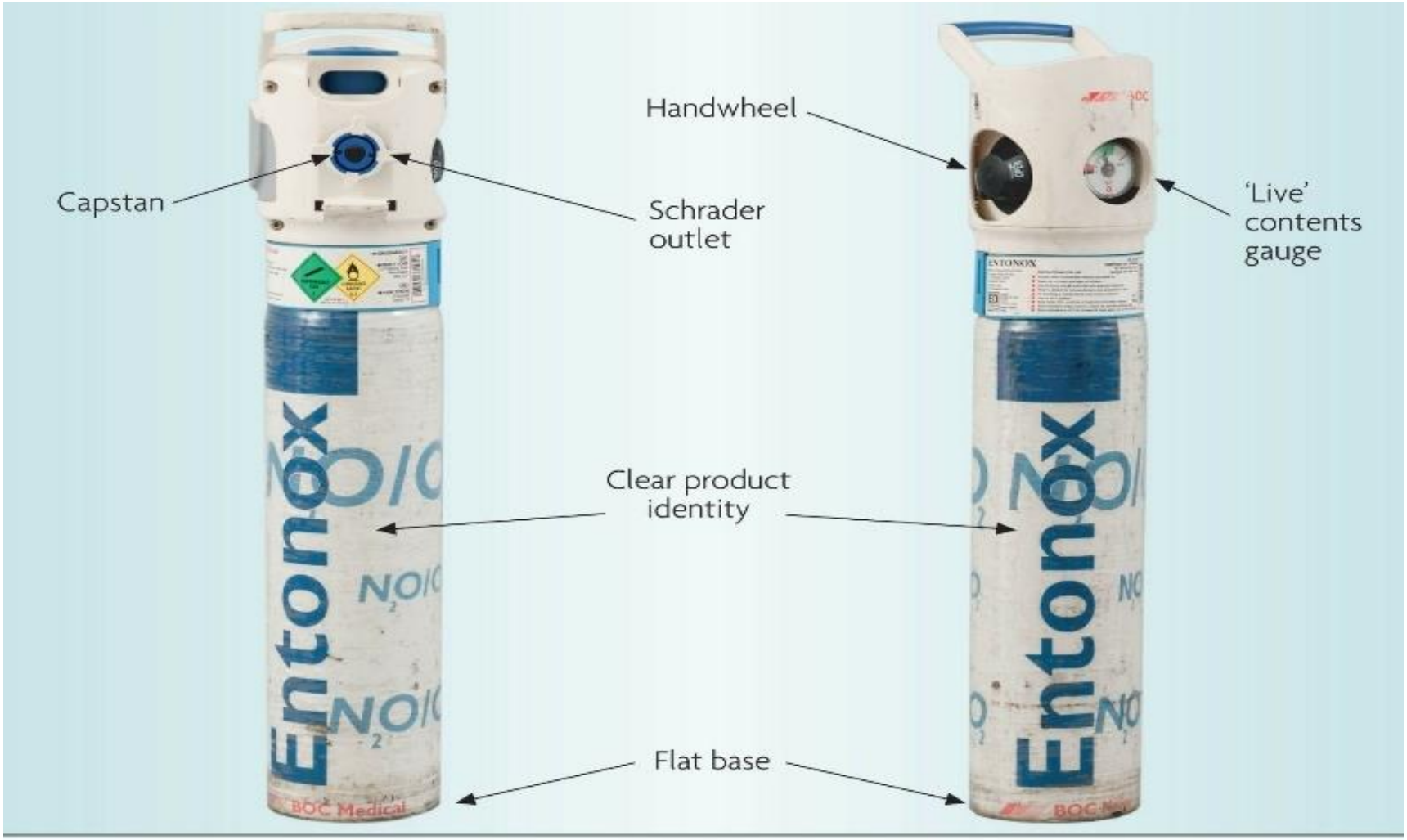
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Entonox

- Entonox is a combination of 50% nitrous oxide and 50% oxygen.
- It is an inhaled, self administered gas that is quick acting but also has a rapid reduction in analgesic effect when removed
- When the temperature of the gases falls below -6c the two gases separate.
- Before use you must invert the cylinder a number of times to mix the two gases if it has been stored at the above temperature

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1. Ensure the demand valve is clean and ready for use.



2. Fit a new microbial filter and mouthpiece to the demand valve. They are single use.

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3. Insert the probe on the hose connected to the demand valve into the Schrader outlet on the Entonox cylinder. Push firmly to ensure the probe clicks into place.



4. Slowly turn the cylinder on by rotating the handwheel anti-clockwise until fully open. Do not use excessive force.

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5. Check the demand valve is operative by pushing the 'test button'. You should hear the gas flowing.



6. Check for leaks, which may be indicated by a hissing sound. If you suspect that you have a leak, turn off the cylinder and check the equipment is properly connected. Turn on the cylinder and recheck for leaks. If the leak continues, turn off and remove the

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Entonox

- Indications
- Contra-indications
- Cautions
- Side Effects
- Dosage
- Routes of administration

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Any Questions?

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